PHP MYSQL

INTRODUCTION

What is MySQL?

✓MySQL is a database.

✓The data in MySQL is stored in database objects called tables.

✓A table is a collections of related data entries and it consists of columns and rows.

✓Databases are useful when storing information categorically.

✓A company may have a database with the following tables: *"Employees", "Products", "Customers" and "Orders".*

INTRODUCTION

Queries?

✓A query is a question or a request.

✓With MySQL, we can query a database for specific information and have a recordset returned.

EXAMPLE:

```
SELECT LastName FROM Persons
```

HOW TO CREATE A CONNECTION?

SYNTAX

```
mysql_connect(servername,username,password);
```

Parameter	Description
servername	Optional. Specifies the server to connect to. Default value is "localhost:3306"
username	Optional. Specifies the username to log in with. Default value is the name of the user that owns the server process
password	Optional. Specifies the password to log in with. Default is ""

HOW TO CREATE A CONNECTION?

EXAMPLE

dbconn.php

```
<?php
$con = mysql_connect("localhost","peter","abc123");
if (!$con)
  {
  die('Could not connect: ' . mysql_error());
  }

// some code
?>
```

change usernama & password to your own server web hosting.

HOW TO CLOSE A CONNECTION?

SYNTAX

```
<?php
$con = mysql_connect("localhost","peter","abc123");
if (!$con)
  {
  die('Could not connect: ' . mysql_error());
  }

// some code

mysql_close($con);
?>
```

HOW TO CREATE A DATABASE?

✓A database holds one or multiple tables.

✓**CREATE DATABASE** statement is used to create a database in MySQL.

SYNTAX

```
CREATE DATABASE database_name
```

To get PHP to execute the statement above we must use the mysql_query() function. This function is used to send a query or command to a MySQL connection.

HOW TO CREATE A DATABASE?

Example creates a database called "my_db":

SYNTAX

create_db.php

```
<?php
$con = mysql_connect("localhost","peter","abc123");
if (!$con)
  {
        die('Could not connect: ' . mysql_error());
  }

if (mysql_query("CREATE DATABASE my_db",$con))
  {
        echo "Database created";
  }
else
  {
        echo "Error creating database: " . mysql_error();
  }

mysql_close($con);
?>
```

change my_db to your own server db e.g: db_nafbti

HOW TO CREATE A TABLE?

Example creates a table named "Persons", with three columns

SYNTAX

create_tbl.php

```
<?php
//Create connection
// Create database
// Create table
mysql_select_db("my_db", $con);
$sql = "CREATE TABLE Persons
(
FirstName varchar(15),
LastName varchar(15),
Age int
)";

// Execute query
mysql_query($sql,$con);

mysql_close($con);
?>
```

Add the CREATE TABLE statement to the mysql_query() function to execute the command.

Important: A database must be selected before a table can be created. The database is selected with the mysql_select_db() function.

change my_db to your own server db e.g: db_nafbti

HOW TO CREATE A TABLE?

Primary Keys and Auto Increment Fields

```
$sql = "CREATE TABLE Persons
(
personID int NOT NULL AUTO_INCREMENT,
PRIMARY KEY(personID),
FirstName varchar(15),
LastName varchar(15),
Age int
)";

mysql_query($sql,$con);
```

Example creates a table named "Persons", with three columns

INSERT DATA TO DATABASE TABLE

SYNTAX

insert_data.php

```
<?php
$con = mysql_connect("localhost","peter","abc123");
if (!$con)
  {
  die('Could not connect: ' . mysql_error());
  }

mysql_select_db("my_db", $con);

mysql_query("INSERT INTO Persons (FirstName, LastName, Age)
VALUES ('Peter', 'Griffin', '35')");

mysql_query("INSERT INTO Persons (FirstName, LastName, Age)
VALUES ('Glenn', 'Quagmire', '33')");

mysql_close($con);
?>
```

INSERT DATA FROM A FORM INTO A DATABASE TABLE

EXAMPLE

frm_insert.html

```
<html>
<body>

        <form action="insert.php" method="post">
        Firstname: <input type="text" name="firstname" />
        Lastname: <input type="text" name="lastname" />
        Age: <input type="text" name="age" />
        <input type="submit" />
        </form>

</body>
</html>
```

Create an HTML form that can be used to add new record for PERSONS table

INSERT DATA FROM A FORM INTO A DATABASE TABLE

EXAMPLE

insert.php

```
<?php
//create connection

mysql_select_db("my_db", $con);

$sql="INSERT INTO Persons (FirstName, LastName, Age)
VALUES ('$_POST[firstname]','$_POST[lastname]','$_POST[age]')";

if (!mysql_query($sql,$con))
  {
  die('Error: ' . mysql_error())
  }
echo "1 record added";

mysql_close($con)
?>
```

- When a user clicks the submit button in the HTML form in the example above, the form data is sent to "insert.php".
- The "insert.php" file connects to a database, and retrieves the values from the form with the PHP $_POST variables.
- Then, the mysql_query() function executes the INSERT INTO statement, and a new record will be added to the "Persons" table.

SELECT DATA FROM A DATABASE TABLE

EXAMPLE

select.php

```
<?php
$con = mysql_connect("localhost","peter","abc123");
if (!$con)
  {
  die('Could not connect: ' . mysql_error());
  }

mysql_select_db("my_db", $con);

$result = mysql_query("SELECT * FROM Persons");

while($row = mysql_fetch_array($result))
  {
  echo $row['FirstName'] . " " . $row['LastName'];
  echo "<br />";
  }

mysql_close($con);
?>
```

- stores the data returned by the mysql_query() function in the $result variable.
- use the mysql_fetch_array() function to return the first row from the recordset as an array
- Each call to mysql_fetch_array() returns the next row in the recordset.
- while loop loops through all the records in the recordset. To print the value of each row, we use the PHP $row variable ($row['FirstName'] and $row['LastName']).

SELECT DATA FROM A DATABASE TABLE

Display the Result in an HTML Table.

The following example selects the same data as the example above, but will display the data in an HTML

```
<?php
$con = mysql_connect("localhost","peter","abc123");
if (!$con)
{
die('Could not connect: ' . mysql_error());
}
mysql_select_db("my_db", $con);
$result = mysql_query("SELECT * FROM Persons");
echo "<table border='1'>
<tr>
<th>Firstname</th>
<th>Lastname</th>
</tr>";
while($row = mysql_fetch_array($result))
{
echo "<tr>";
echo "<td>" . $row['FirstName'] . "</td>";
echo "<td>" . $row['LastName'] . "</td>";
echo "</tr>";
}
echo "</table>";
mysql_close($con);
?>
```

The Where Clause

- The WHERE clause is used to filter records.
- The WHERE clause is used to extract only those records that fulfill a specified criterion.

Syntax

```
SELECT column_name(s)
FROM table_name
WHERE column_name operator value
```

Example

The following example selects all rows from the "Persons" table where "FirstName='Peter':

where.php

```
<?php
//connection to server
$con = mysql_connect("localhost","root","");
if (!$con)
    {
        die('Could not connect: ' . mysql_error());
        }

//select the dbase
mysql_select_db("my_db", $con);

$result = mysql_query("SELECT * FROM Persons where firstname = 'Peter'");
while($row = mysql_fetch_array($result))
     {
         echo $row['FirstName'] . " " . $row['LastName'];
         echo "<br/>";
         }

mysql_close($con);

?>
```

The ORDER BY Keyword

- The ORDER BY keyword is used to sort the data in a recordset.
- The ORDER BY keyword sort the records in ascending order by default.
- If you want to sort the records in a descending order, you can use the DESC keyword.

Syntax

```
SELECT column_name(s)
FROM table_name
ORDER BY column_name(s) ASC|DESC
```

Example
The following example selects all the data stored in the "Persons" table, and sorts the result by the "Age" column:

where.php

```
<?php
//connection to server
$con = mysql_connect("localhost","root","");
if (!$con)
    {
        die('Could not connect: ' . mysql_error());
        }

//select the dbase
mysql_select_db("my_db", $con);

$result = mysql_query("SELECT * FROM Persons ORDER BY age");
while($row = mysql_fetch_array($result))
     {
         echo $row['FirstName'] . " " . $row['LastName'];
         echo "<br/>";
         }

mysql_close($con);

?>
```

The ORDER BY Keyword (ASC|DESC)

- The sorting can be charged from ascending to descending by adding ASC or DESC:

```
$result = mysql_query("SELECT * FROM Persons ORDER BY age DESC");
```

The Where Clause (SELECT DATA FROM A FORM)

- The WHERE clause is used to filter records.
- The WHERE clause is used to extract only those records that fulfill a specified criterion.

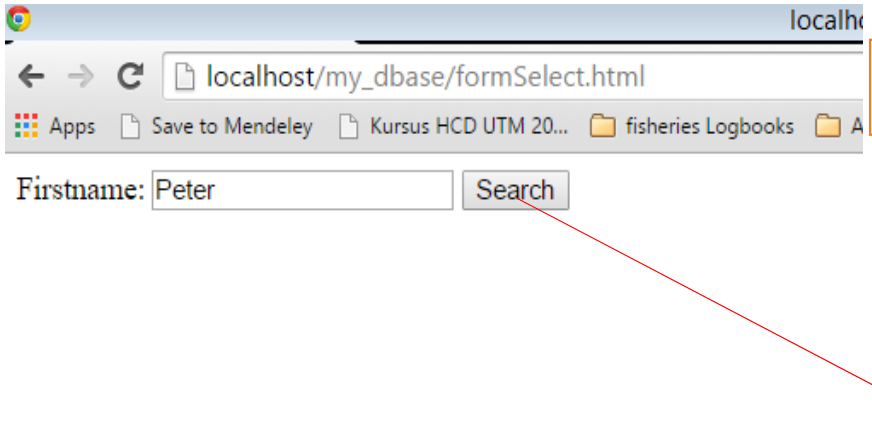
Syntax

```
SELECT column_name(s)
FROM table_name
WHERE column_name operator value
```

The Where Clause (SELECT DATA FROM A FORM)

Example

The following example selects all rows from the "Persons" table where the user input "FirstName" from the textbox.



```
<html>
<body>
<form action="selectfrm.php" method="post">
Firstname: <input type="text" name="firstname" />
<input type="submit" value="Search" />
</form>
</body>
</html>
```

frm_select.html

```
<?php
//connection to server
$con = mysql_connect("localhost","root","");
if (!$con)
  {
        die('Could not connect: ' . mysql_error());
        }

//select the dbase
mysql_select_db("my_db", $con);

$result = mysql_query("SELECT * FROM Persons where
FirstName=('$_POST[firstname]')");
while($row = mysql_fetch_array($result))
  {
        echo $row['FirstName'] . " " . $row['LastName'];
        echo "<br/>";
        }

mysql_close($con);

?>
```

selectfrm.php

DELETE THE DATA (DELETE DATA FROM A FORM)

- When deleting a record, you can use the unique AutoNumber field in the database.
- Using this unique identifier ensures that you only delete one record.
- REMEMBER! Once you have deleted a record, it is gone and cannot be restored.

DELETE THE DATA
(DELETE DATA FROM A FORM)

Syntax

```
DELETE FROM TableName WHERE condition
```

Example

The following example delete the rows from the "Persons" table where the user input "FirstName" from the textbox.

Firstname: [ ] Delete

```
<html>
<body>
<form action="deletefrm.php" method="post">
Firstname: <input type="text" name="firstname" />
<input type="submit" value="Delete" />
</form>
</body>
</html>
```

frm_delete.html

```
<?php
//connection to server
$con = mysql_connect("localhost","root","");
if (!$con)
  {
        die('Could not connect: ' . mysql_error());
        }

//select the database
mysql_select_db("my_db", $con);

// The SQL statement that deletes the record
$result = "DELETE FROM Persons where
FirstName=('$_POST[firstname]')";
mysql_query($result);

// Close the database connection
mysql_close($con);

?>
```

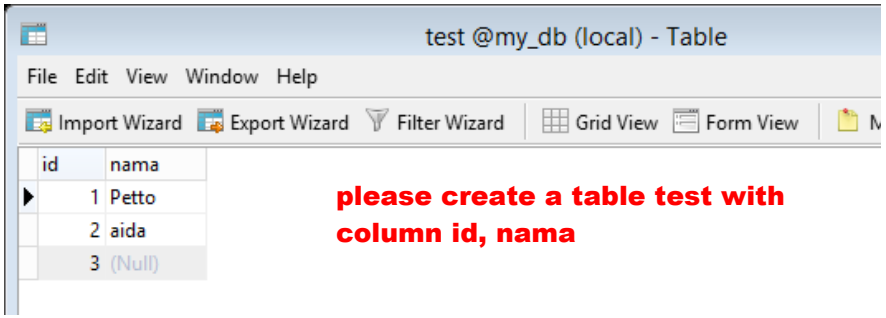
deletefrm.php

UPDATE THE DATA

Syntax

UPDATE TableName SET TableColumn='value'
WHERE condition

CREATE 2nd TABLE name as test



please create a table test with column id, nama

update.php

```
<?php
//connection to server
$con = mysql_connect("localhost","root","");
if (!$con)
  {
        die('Could not connect: ' . mysql_error());
        }

//select the database
mysql_select_db("my_db", $con);

// The SQL statement that deletes the record
$result = "update test set nama='Petto' where id=1";
mysql_query($result);

// Close the database connection
mysql_close($con);

?>
```

create_tbl_test.php

create_tbl_test.php

```php
<?php

$con = mysql_connect("localhost","nafbti","8118");
if (!$con)
{
die('Could not connect: ' . mysql_error());
}

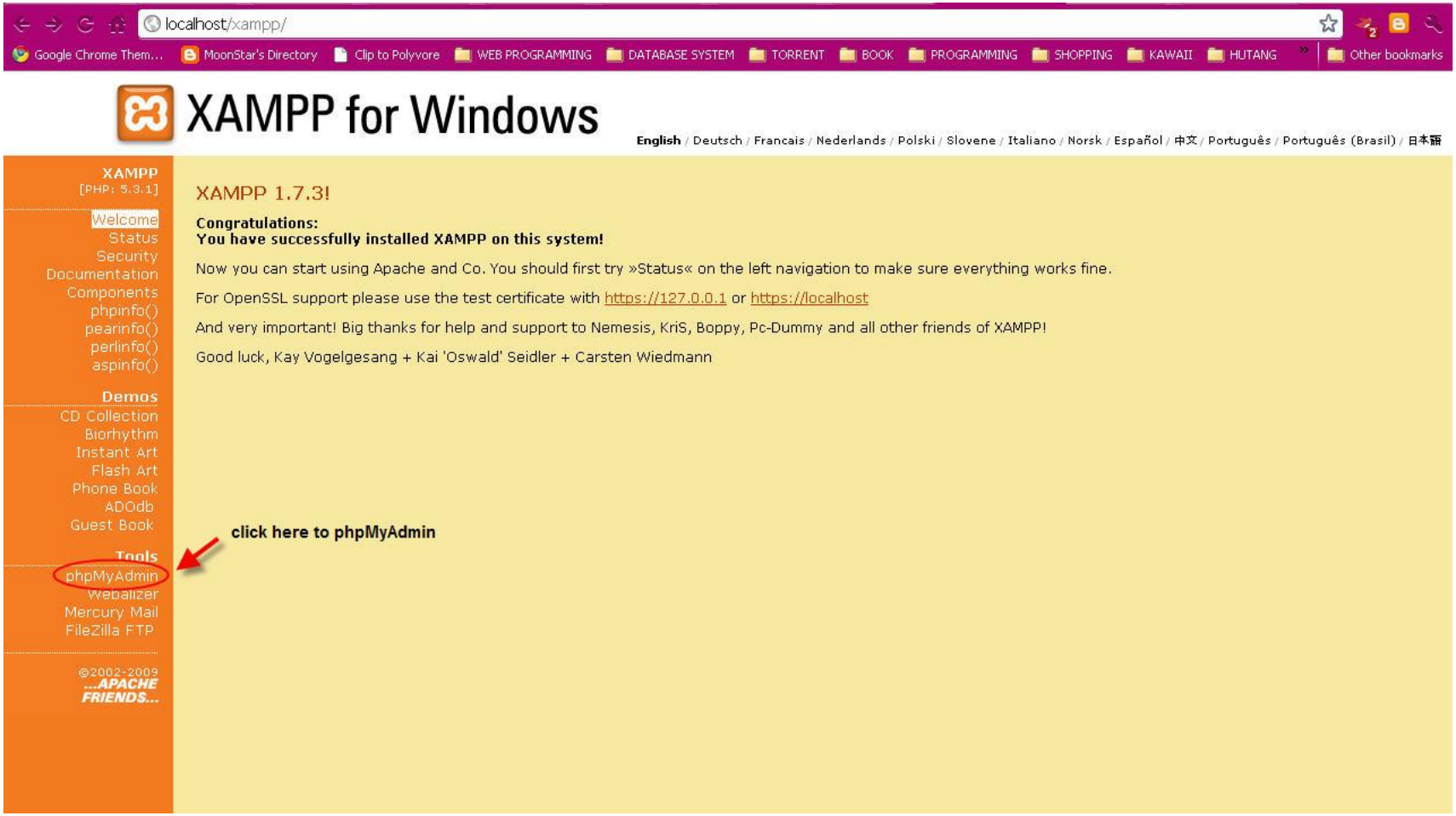

// query create table in my_db1
mysql_select_db("db_nafbti", $con);
$sql = "CREATE TABLE test
(
Name varchar(15),
id int
)";

// Execute query
mysql_query($sql,$con);

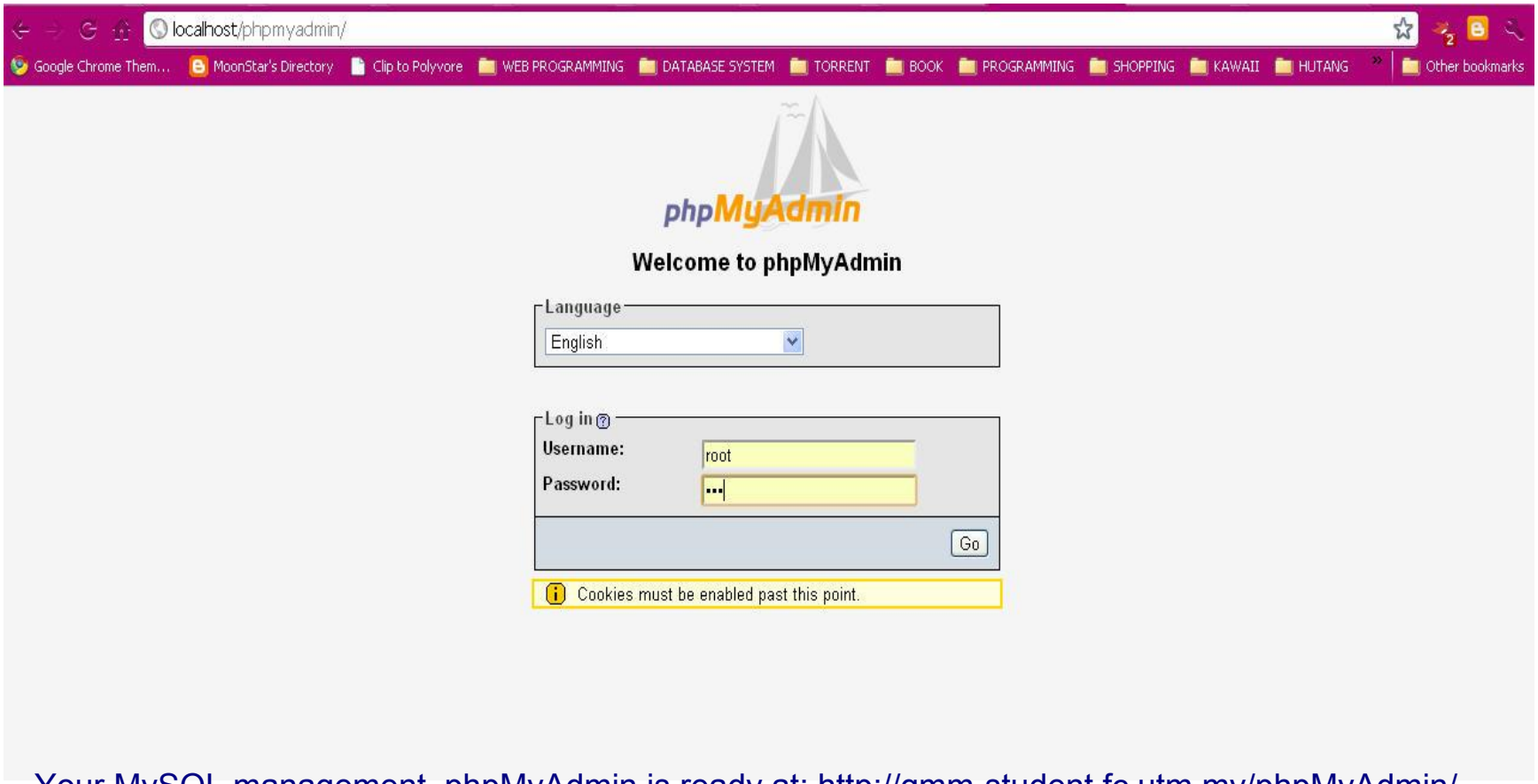
mysql_close($con);
?>
```

Change server username, password and database to your own server.

STEP BY STEP TO CONNECT WITH PHPMYADMIN

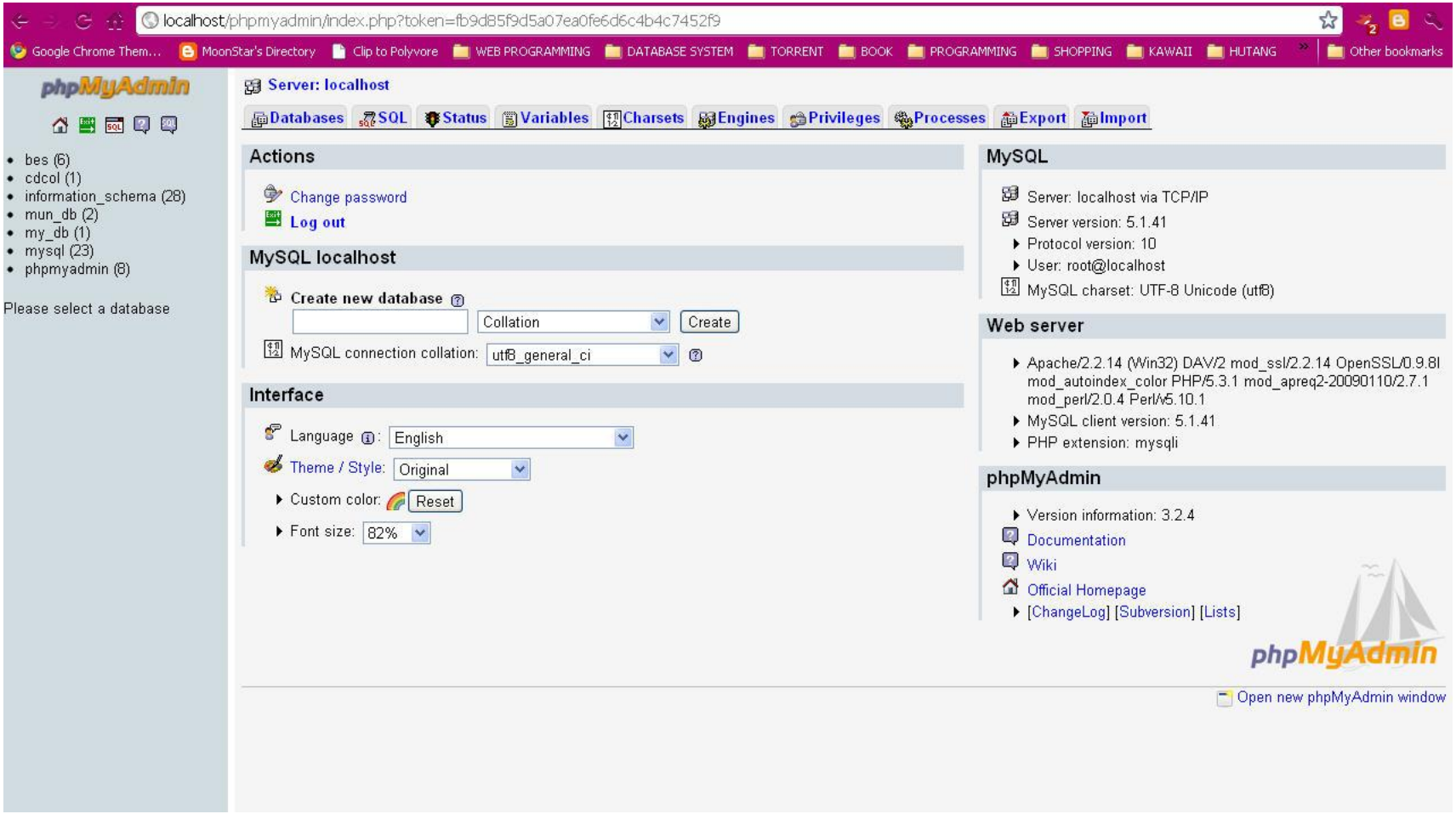


STEP BY STEP TO CONNECT WITH PHPMYADMIN

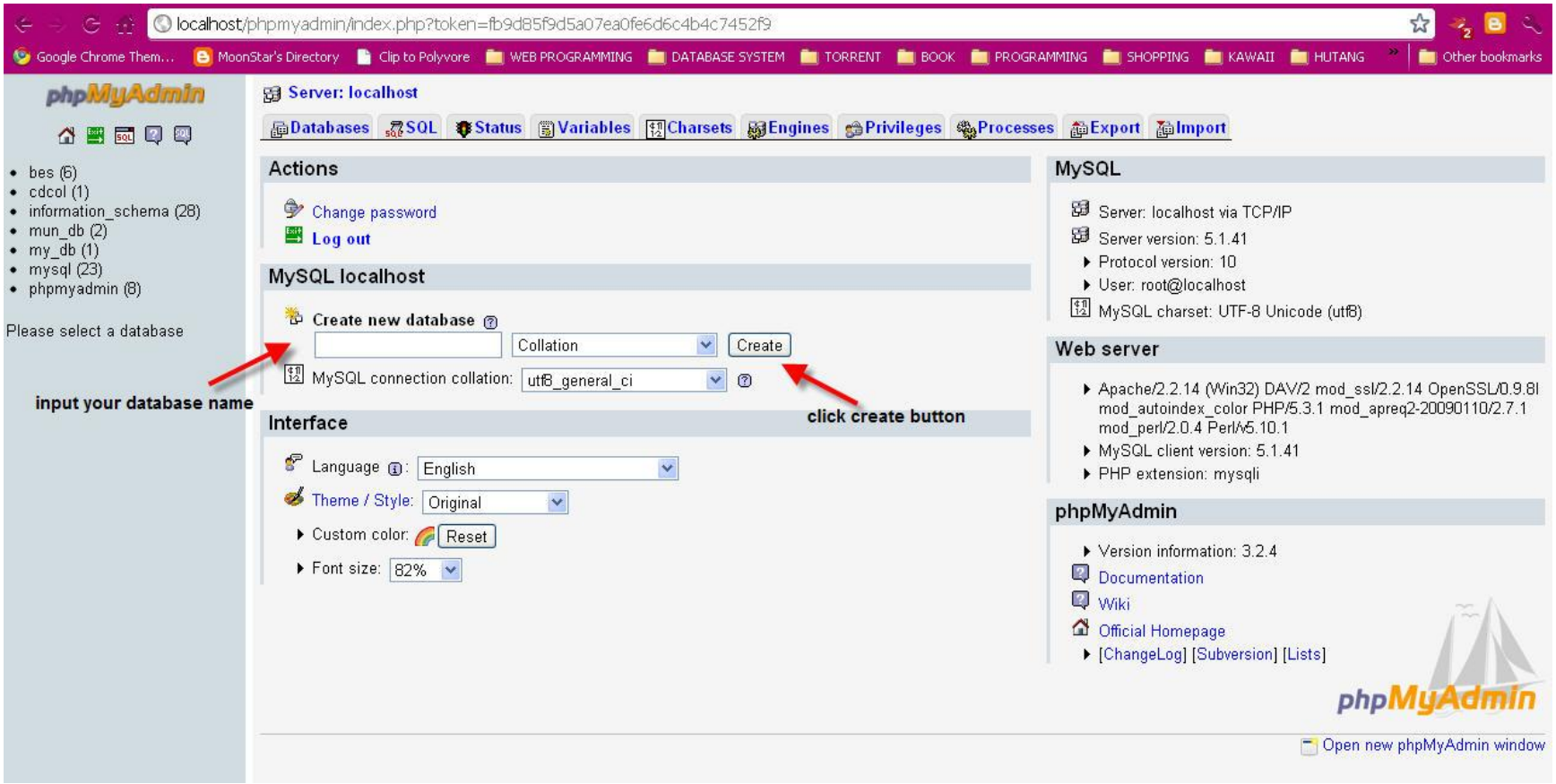


Your MySQL management, phpMyAdmin is ready at: http://gmm-student.fc.utm.my/phpMyAdmin/

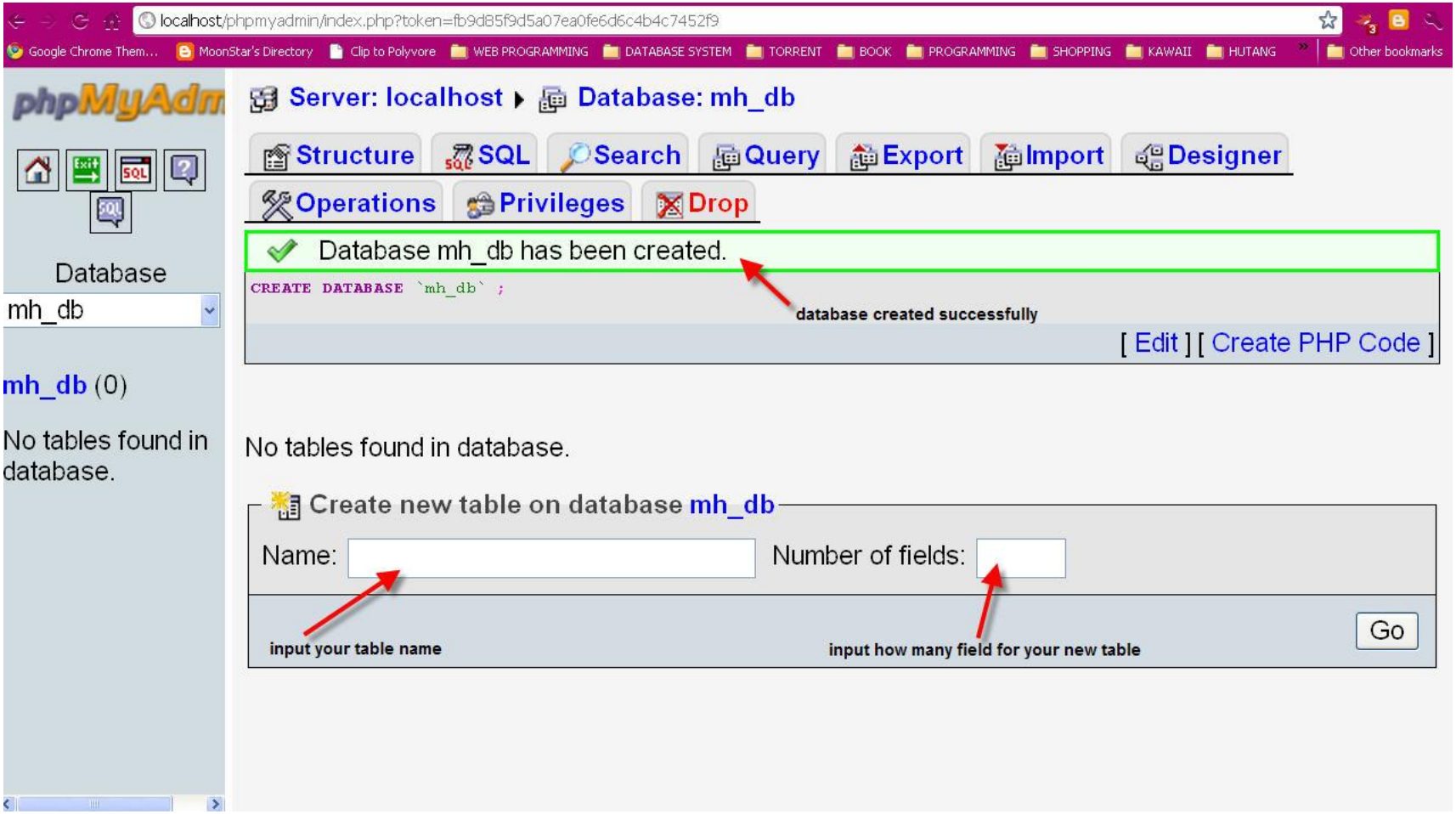
STEP BY STEP TO CONNECT WITH PHPMYADMIN



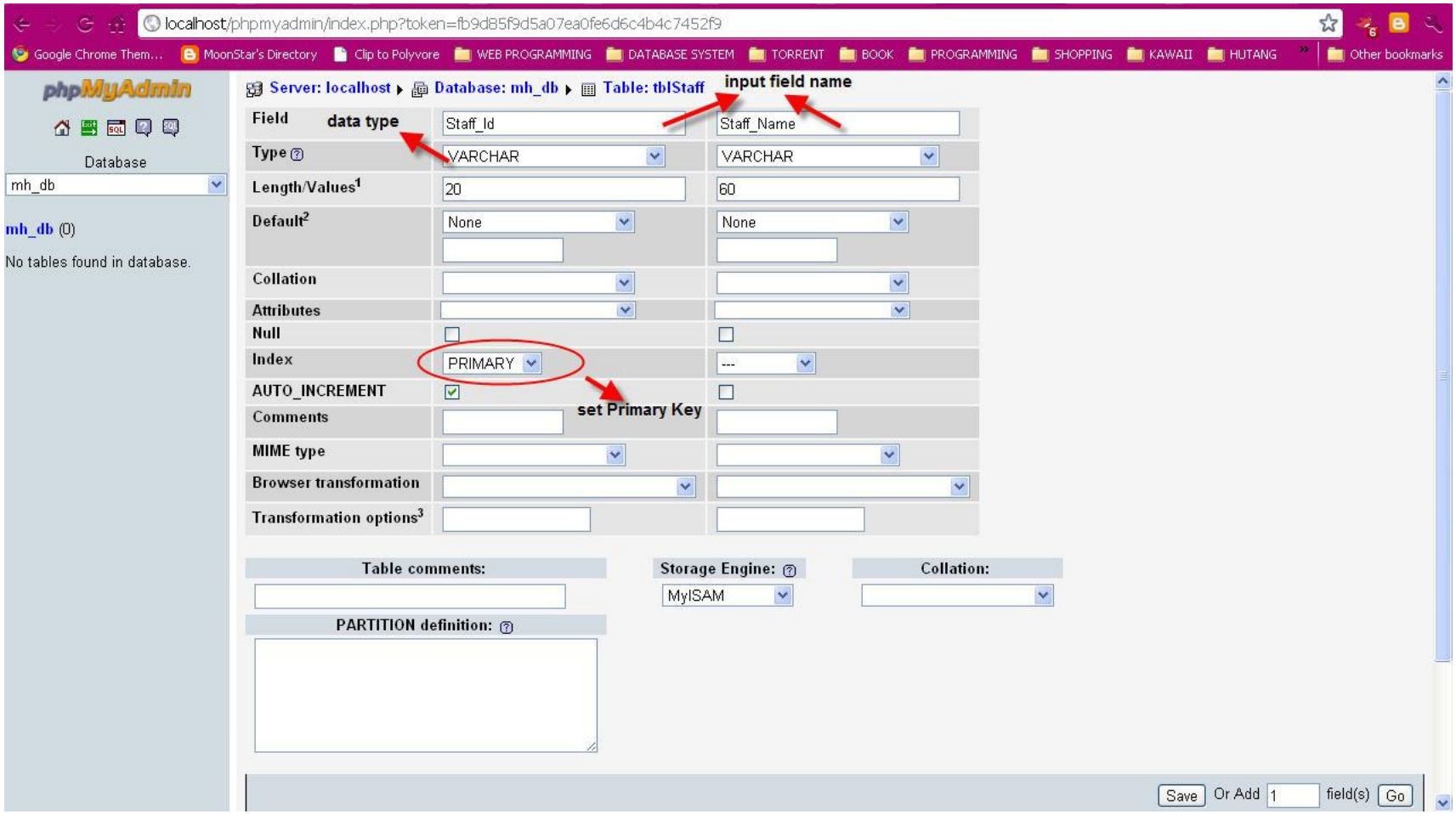
STEP BY STEP TO CONNECT WITH PHPMYADMIN



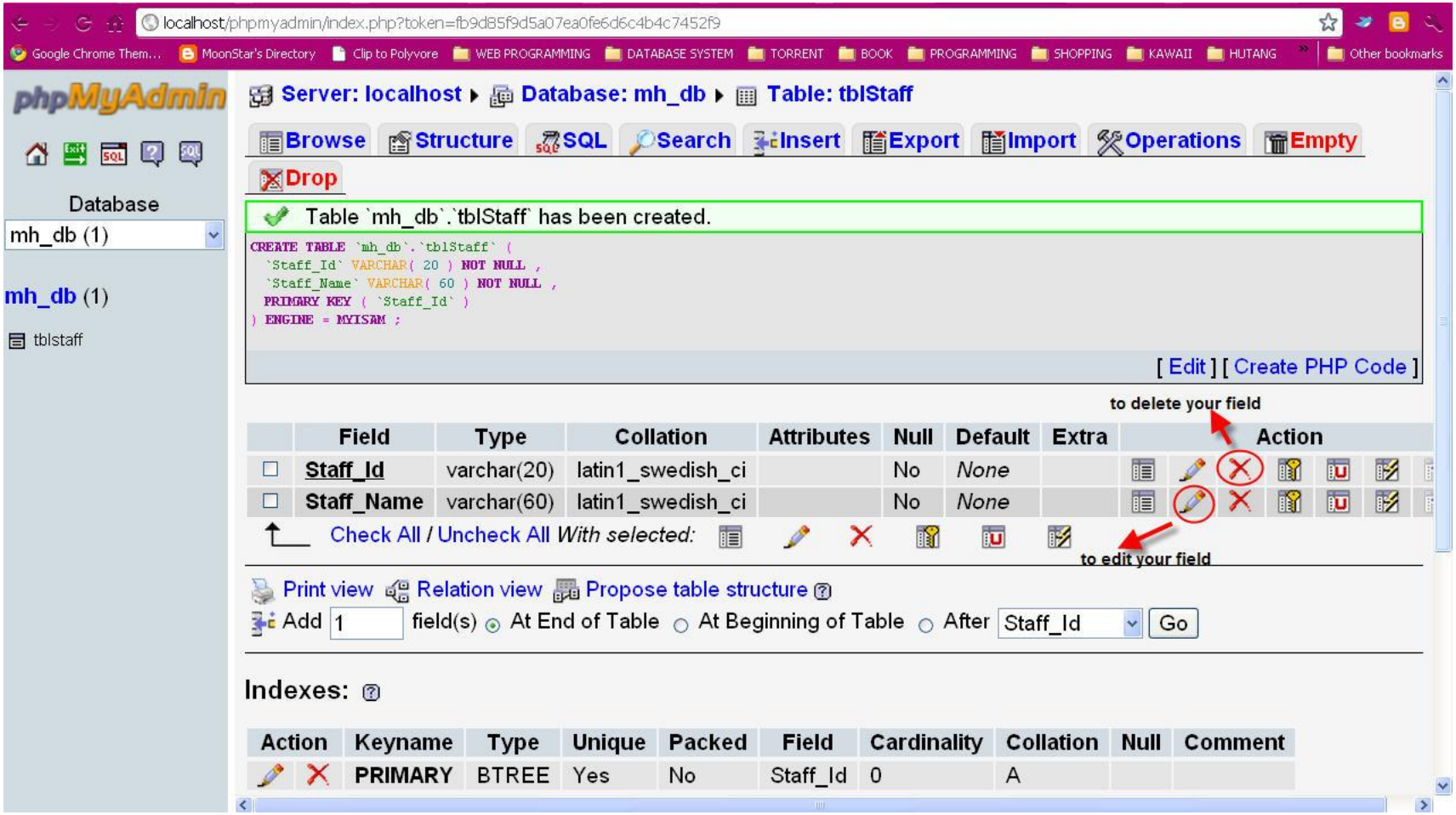
STEP BY STEP TO CONNECT WITH PHPMYADMIN



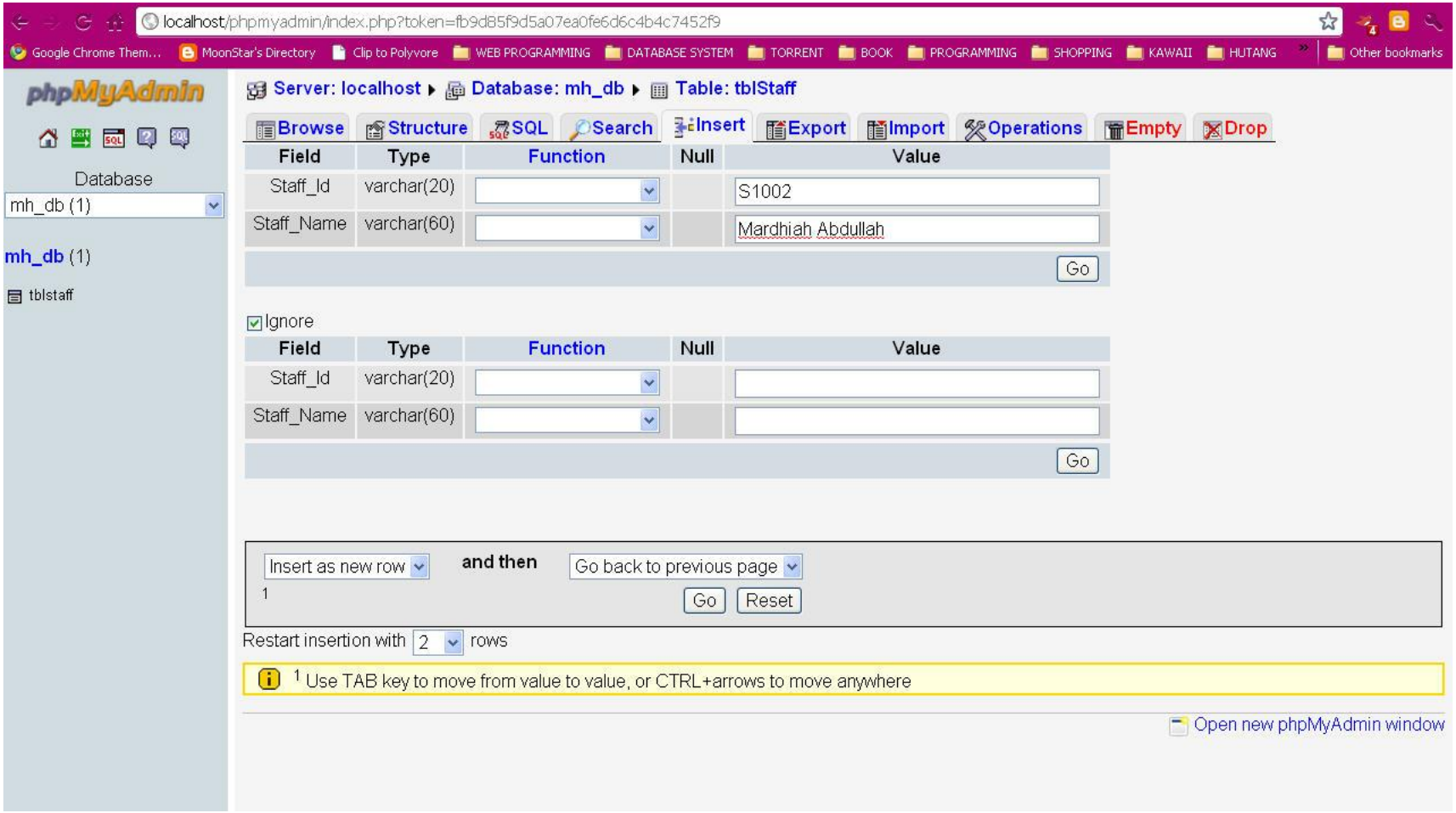
STEP BY STEP TO CONNECT WITH PHPMYADMIN



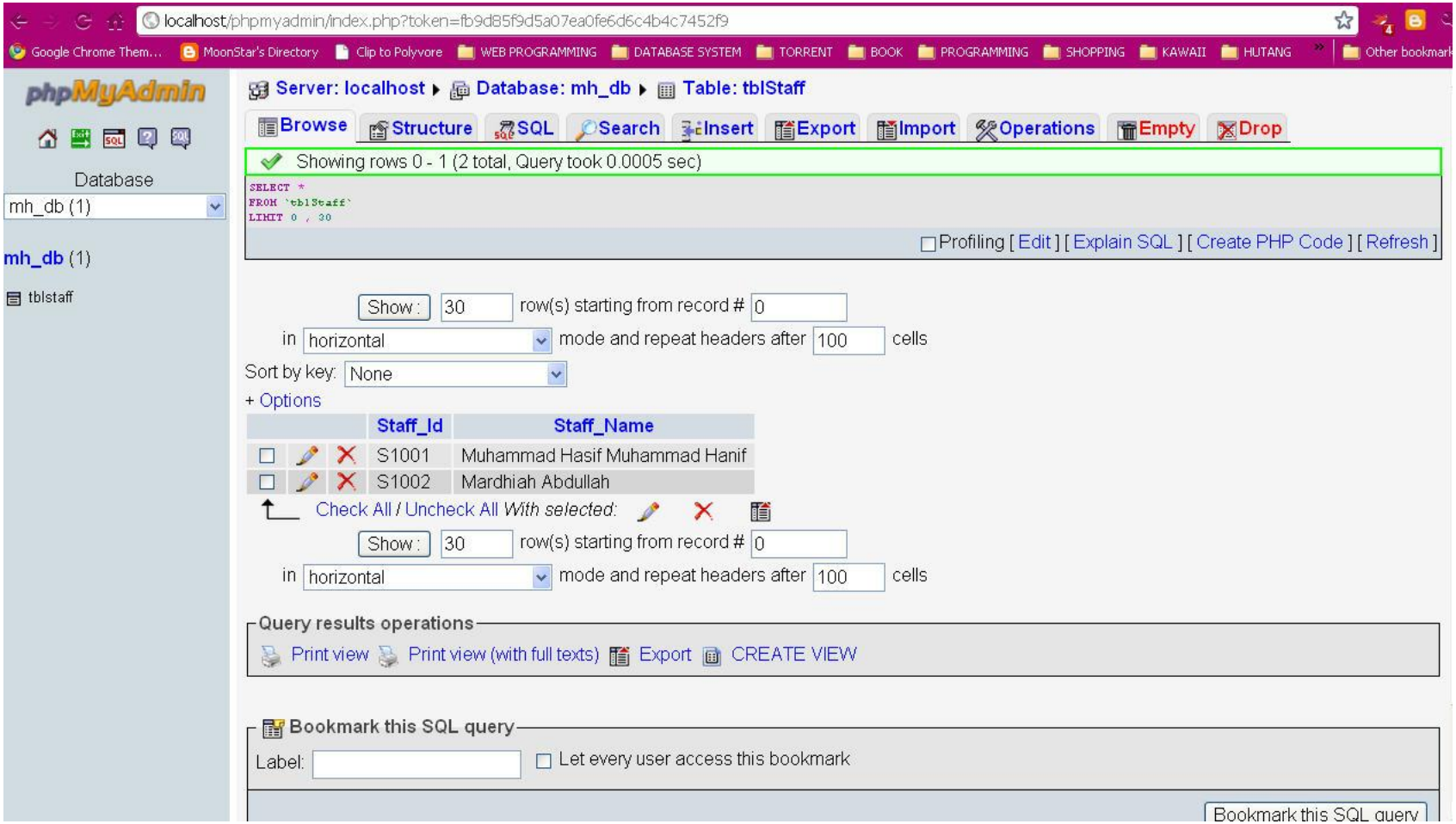
STEP BY STEP TO CONNECT WITH PHPMYADMIN



STEP BY STEP TO CONNECT WITH PHPMYADMIN



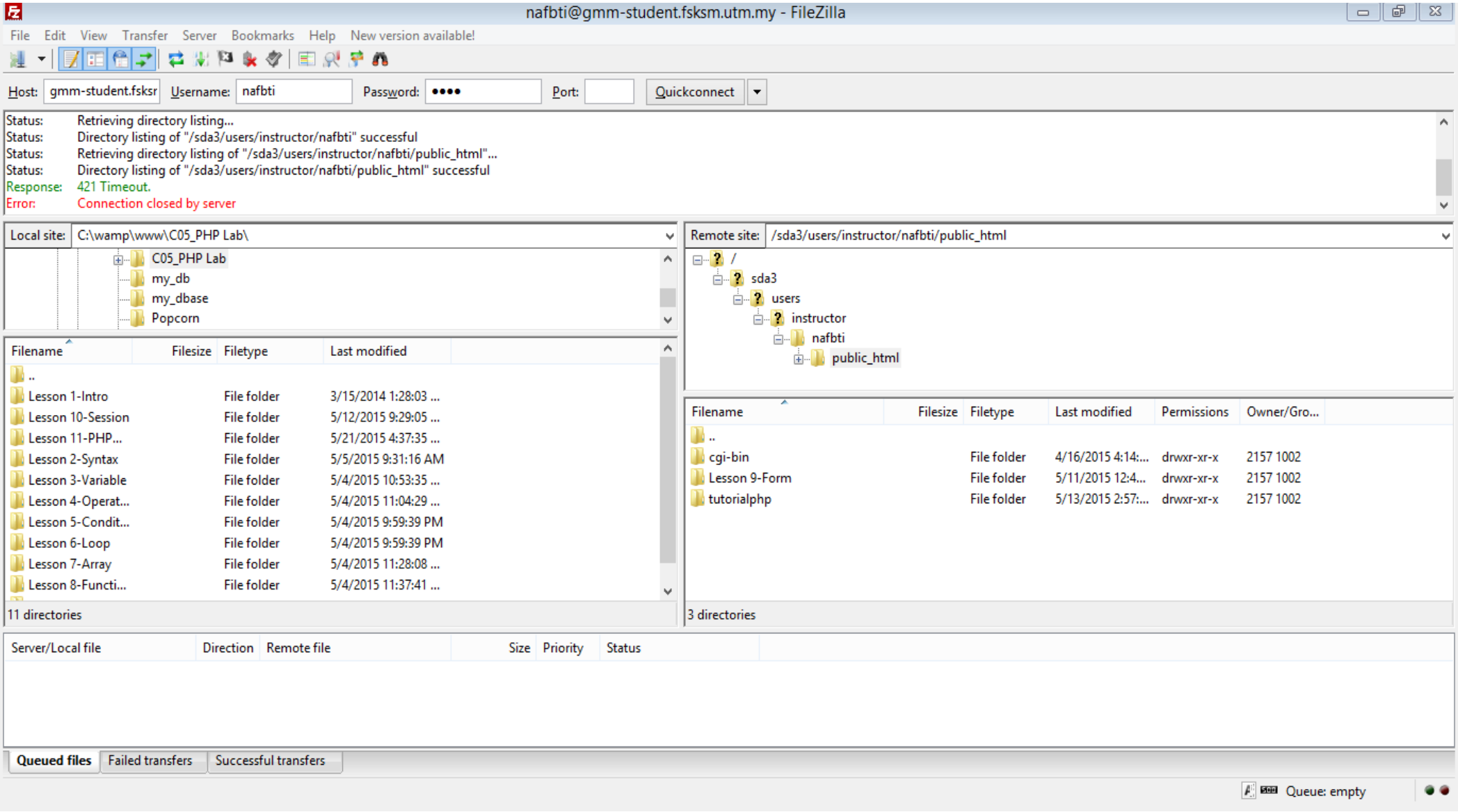
STEP BY STEP TO CONNECT WITH PHPMYADMIN



Server gmm-student

- Run Fillzila
- Key in the information below:
  - Host name : gmm-student.fc.utm.my
  - Username : your loginname
  - Password : your matric no (lowercase)
- Click QuickConnect
- Access your Web site on the server gmm =>

  http://gmm-student.fc.utm.my/~yourlogin/


nafbti@gmm-student.fsksm.utm.my - FileZilla
File Edit View Transfer Server Bookmarks Help New version available!
Host: gmm-student.fsksr Username: nafbti Password: •••• Port: Quickconnect
Status: Retrieving directory listing...
Status: Directory listing of "/sda3/users/instructor/nafbti" successful
Status: Retrieving directory listing of "/sda3/users/instructor/nafbti/public_html"...
Status: Directory listing of "/sda3/users/instructor/nafbti/public_html" successful
Response: 421 Timeout.
Error: Connection closed by server
Local site: C:\wamp\www\C05_PHP Lab\
C05_PHP Lab
my_db
my_dbase
Popcorn
Filename Filesize Filetype Last modified
..
Lesson 1-Intro File folder 3/15/2014 1:28:03 ...
Lesson 10-Session File folder 5/12/2015 9:29:05 ...
Lesson 11-PHP... File folder 5/21/2015 4:37:35 ...
Lesson 2-Syntax File folder 5/5/2015 9:31:16 AM
Lesson 3-Variable File folder 5/4/2015 10:53:35 ...
Lesson 4-Operat... File folder 5/4/2015 11:04:29 ...
Lesson 5-Condit... File folder 5/4/2015 9:59:39 PM
Lesson 6-Loop File folder 5/4/2015 9:59:39 PM
Lesson 7-Array File folder 5/4/2015 11:28:08 ...
Lesson 8-Functi... File folder 5/4/2015 11:37:41 ...
11 directories
Remote site: /sda3/users/instructor/nafbti/public_html
/
sda3
users
instructor
nafbti
public_html
Filename Filesize Filetype Last modified Permissions Owner/Gro...
..
cgi-bin File folder 4/16/2015 4:14:... drwxr-xr-x 2157 1002
Lesson 9-Form File folder 5/11/2015 12:4... drwxr-xr-x 2157 1002
tutorialphp File folder 5/13/2015 2:57:... drwxr-xr-x 2157 1002
3 directories
Server/Local file Direction Remote file Size Priority Status
Queued files Failed transfers Successful transfers
Queue: empty

THE END